

DataBase System:

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Lab Task:

1.

```
> db.books.countDocuments({ author: "George Orwell" })  
< 2  
libraryDB>
```

2.

```
> db.books.find({ publication_year: { $gt: 2000 } })  
< {  
  _id: ObjectId('69f62738965ccac2a72c51e0'),  
  title: 'The Da Vinci Code',  
  author: 'Dan Brown',  
  genre: 'Thriller',  
  publication_year: 2003,  
  rating: 4.1  
}
```

3.

```
> db.books.updateOne(  
  { title: "The Catcher in the Rye"},  
  { $set: {genre: "Classic Fiction"}}  
)  
< {  
  acknowledged: true,  
  insertedId: null,  
  matchedCount: 1,  
  modifiedCount: 1,  
  upsertedCount: 0  
}
```

4.

```
> db.books.updateMany(
  {},
  { $inc: { rating: 0.5}}
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 5,
  modifiedCount: 5,
  upsertedCount: 0
}
```

5.

```
> db.books.find({
  $or: [
    { title: { $regex: "Great", $options: "i" } },
    { author: { $regex: "Great", $options: "i" } }
  ]
})
< {
  _id: ObjectId('69f62738965ccac2a72c51de'),
  title: 'The Great Gatsby',
  author: 'F. Scott Fitzgerald',
  genre: 'Classic',
  publication_year: 1925,
  rating: 4.9
}
```

6.

```
> db.books.find().sort({ publication_year: -1 })
< {
  _id: ObjectId('69f62738965ccac2a72c51e0'),
  title: 'The Da Vinci Code',
  author: 'Dan Brown',
  genre: 'Thriller',
  publication_year: 2003,
  rating: 4.6
}
{
  _id: ObjectId('69f62738965ccac2a72c51df'),
  title: 'The Catcher in the Rye',
  author: 'J.D. Salinger',
  genre: 'Classic Fiction',
  publication_year: 1951,
  rating: 4.5
}
{
  _id: ObjectId('69f62738965ccac2a72c51dc'),
  title: '1984',
  author: 'George Orwell',
  genre: 'Dystopian',
  publication_year: 1949,
  rating: 5
}
{
  _id: ObjectId('69f62738965ccac2a72c51dd'),
  title: 'Animal Farm',
```

7.

```
> db.books.aggregate([
  {
    $group: {
      _id: "$genre",
      avgYear: { $avg: "$publication_year" }
    }
  }
])
< {
  _id: 'Classic Fiction',
  avgYear: 1951
}
{
  _id: 'Political Satire',
  avgYear: 1945
}
{
  _id: 'Dystopian',
  avgYear: 1949
}
{
  _id: 'Classic',
  avgYear: 1925
}
{
  _id: 'Thriller',
```

8.

```
> db.books.updateMany(
  {},
  { $set: { available: true } }
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 5,
  modifiedCount: 5,
  upsertedCount: 0
}
```

9.

```
> db.books.deleteMany({ publication_year: { $lt: 1950 } })
< {
  acknowledged: true,
  deletedCount: 3
}
```

10.

```
> db.books.distinct("genre")
< [ 'Classic Fiction', 'Thriller' ]
```